

Date: June 2026 | Rev: v2.0

MCU: U1 – ESP32-C3-MINI-1
3.3V supply | GND | EN pin pulled HIGH via 10k | I02 strapping to GND
USB-C D+ D- to ESP32-C3 native USB pins
I2C: SDA=GPIO8, SCL=GPIO9 | SPI: CLK=GPIO4, MOSI=GPIO6, MISO=GPIO2

I2C SENSORS (SDA=GPIO8, SCL=GPIO9)
U2: Sensing SH740 (addr 0x3D) Temp/Humidity VDD=3.3V | 100nF decoupling
U3: Microchip ATECC608A | Addr 0x00 | HW Crypto ECC-P256 | 100nF decoupling
U4: ST ST25DV04K | Addr 0x53 | NFC tag + I2C EEPROM | RF energy harvesting
U9: Maxim MAX17048 | Addr 0x36 | LiPo Fuel Gauge | SDA/SCL shared bus

SPI BUS (CLK=GPIO4, MOSI=GPIO6, MISO=GPIO2)
U5: ADXL345 Accelerometer | CS=GPIO5 | ±16g range, full resolution, 100Hz ODR
U6: SX1262 LoRa Module | CS=GPIO7, BUSY=GPIO3, DIO1=GPIO10, RST=GPIO1
J3: e-ink Display FPC 8-pin | CS=GPIO19, DC=GPIO18, RST=GPIO17, BUSY=GPIO16
B 1.54" 200x200 B/W | SSD1681 driver IC | zero-power image retention

LEDS
D1 LED_RED_GPIO12
D2 LED_GREEN_GPIO13
D3 LED_BLUE_GPIO14
D4 LED_BAT_GPIO15

GPS MODULE (UART)
U7: u-blox MAX-M10S | TX=GPIO21 (ESP RX), RX=GPIO20 (ESP TX) | 9600 baud
Passive antenna or active antenna with LNA bias

POWER SYSTEM
J1: JST-PH 2-pin LiPo connector (1000mAh 3.7V)
J2: USB-C receptacle (TP4056 VBUS + ESP32 D+/D-)
U9: TP4056 LiPo CHG VIN -> LiPo charge | 1A charge current | charge indicator LED
U10: DW01A + FS8205 battery protection IC | overvoltage/overcurrent/reverse
U11: AMS1117-3.3 | LiPo+ -> VBAT -> AMS1117 IN -> 3.3V -> all ICs
D4: LED + R4(330ohm) battery indicator driven from GPIO15

INDICATORS, BUZZER & TAMPER
RGB LED (common cathode) | GPIO12->R1(750)->D1(RED)->GND | GPIO13->R2(330)->D2(GRN)->GND | GPIO14->R3(330)->D3(BLU)->GND
BUZZER: GPIO11->R6(1k)->Q1(2N2222 NPN BASE) | Q1_COLLECTOR->BZ1->3.3V | Q1_EMITTER->GND
TAMPER: GPIO0->R5(10k)->3.3V (pullup) | PCB trace: GPIO0->lid_pad->GND via conductive trace
Lid CLOSED: trace intact, GPIO0=LOW | Lid OPEN: trace broken, GPIO0=HIGH -> TAMPER EVENT

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KiCad E.D.A. 10.0.1	Rev: Id: 1/1

INDICATORS: buzzer & TAMPER

GPIO12->R0(1K)->Q1_2N2222_NPN_BASE | Q1_COLLECTOR->BZ1->3.3V | Q1_EMITTER->GND

TAMPER: GPIO0->R5(10k)->3.3V (pullup) | PCB trace: GPIO0->lid_pad->GND via conductive trace

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